



Understanding Blind and Low Vision Users' Attitudes Towards Spatial Interactions in Desktop Screen Readers

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ABSTRACT

Desktop screen readers as a web navigation mechanism for BLV users are tedious and frozen in time, especially in the face of richer ways of presenting spatial information such as tactile and touch-screen devices. In our work, we consider what it means to create and evaluate systems that can present a similarly rich, spatial interaction mechanism plugged into existing screen reader paradigms. We present a formative study conducted with SpaceNav, a custom screen reader that utilizes spatial input and output to navigate two different web applications. We present results from this study, and discuss a new browser extension we are implementing based on our formative study feedback to more robustly test spatial interactions in the context of real world websites. To close, we describe our goals for evaluating the new web extension in a future study.

KEYWORDS

Blind or low vision users; accessibility; desktop web applications; spatialized layout; screen readers; human-computer interaction

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1 INTRODUCTION

For many blind and low vision (BLV) users, using desktop screen readers to navigate the Web is still a tedious and difficult experience when compared to how sighted users browse the Web. While assistive technologies (ATs) have enabled BLV users to read and interact with websites, as Morris et al. recently described [22], the

experience of consuming visual content via screen readers has remained frozen in time — information is presented linearly, with content read out in a homogenous manner. As a result, BLV website users cannot easily access visual information paradigms such as layout.

Recent research by Potluri et al. shows that many BLV users are interested in learning about visual semantics [26] such as layout, making the problem of conveying visual semantics a ripe area for research. BLV users also report a discrepancy between platforms in conveying visual semantics—mobile screen readers such as iOS's VoiceOver allow users to get a better sense of the positioning of elements via their smartphone or tablet touchscreen, but desktop screen readers offer users less access to visual semantics. Researchers have also recently explored giving BLV users even greater access to spatial layout information via "explore" modes on mobile devices [18] and via tactile printouts [19], yet desktop screen readers continue to lack spatial understanding support.

We seek to evaluate the effects of adding the option of spatial awareness through layout-based interactions into BLV users' daily desktop screen reader workflows. Our research goal is to determine the benefits, drawbacks, and future potential for spatial interactions integrated into desktop screen readers, without adding additional hardware or creating new paradigms. In this work, we integrate spatial interactions into a custom screen reader and conduct a formative study with nine BLV users to understand their attitudes toward spatial input and spatial output. Participants used our spatialized screen reader, SpaceNav, on applications that mimic real world websites. We found that participants felt reduced cognitive load when navigating applications with spatial interactions. They also thought spatial interactions would enable better communication about web pages with sighted people. At the same time, some users found spatialized navigation more challenging to follow compared to traditional screen reader navigation, and they had concerns about the extensibility of the system. We use feedback from the study to develop a web browser extension that allows users to experience any website (such as Google, CNN, Reddit) using spatial interactions in addition to the existing screen reader interactions that they are used to. We close by discussing our goals to evaluate this new web extension in a future study.

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2 BACKGROUND AND RELATED WORK

2.1 Existing Screen Reader Paradigms on Personal Computers and Mobile Devices

Personal computer screen readers and mobile screen readers offer very different experiences for users due to the fact that mobile devices feature touchscreens while desktops and laptops do not. Mobile screen readers such as iOS's VoiceOver [3] have a mode where users can touch their finger over different parts of the screen to hear where user interface elements are located. Comparatively, desktop screen readers present information to users in a linear fashion. Screen readers also treat the arrow keys on the keyboard in different ways: macOS's screen reader VoiceOver uses only the left and right arrow keys in its Reading mode to read the previous and next items, respectively. NVDA [14, 34], an open-source screen reader available for Windows, uses only the up and down arrow keys to read the previous and next items and for line-by-line navigation. It is therefore hard to discern spatial relationships between elements through input keys as there is no standard for directional navigation. Our work bridges the gap between desktop screen readers and mobile screen readers by exploring spatial interaction paradigms that supplement desktop screen reader interaction in a manner that does not require a touchscreen.

2.2 The Importance of Spatial Representations for BLV Users

The importance of learning about visuals in more immersive, spatial ways has long been established by the accessibility research [4, 5, 8–10, 12, 13, 16, 17, 20–23, 26, 28–31, 33, 41] and BLV communities [6, 40]. Researchers have employed printed tactile sheets [4, 17, 21], 3D printed objects [5, 9, 13, 16, 30], and refreshable tactile displays [15, 25, 28] to help BLV users explore specific types of visual content including educational content [9, 13, 16, 21], works of art [1, 29, 31], and graphics [15, 25, 28, 37]. The majority of this prior research has turned to additional devices or hardware setups (tactile displays, 3D printed sheets, touchscreens) to create a spatial understanding of visual content. By contrast, we aim to understand how desktop screen readers can offer users access to spatial information in a manner that does not require specialized hardware and can supplement rather than replace the way that users currently interact with desktop screen readers.

2.3 Nonvisual Interaction with Web Spatial Information

A category of prior research has explored how to promote spatial understanding of Web content specifically. Researchers have primarily explored tactile [19, 19, 32, 32] or additional hardware [24, 27] approaches for representing Web content. A smaller body of work explores auditory approaches to spatial representation. Screen-Track [39] is a tool that turns a website into an auditory overview with different landmark elements (headings, links) represented as various sounds (a dog barking, water droplets), played spatially and concurrently. Wang [38] investigates a screen reader prototype that suffixes screen reader speech with sine tone beeps that symbolize the 2D location of the user's cursor moving. Similarly, some existing

screen readers incorporate spatial audio to an extent via positional audio cues for certain operating system-level controls [2, 34].

Unlike this previous work, we explore the concept of rendering *all* screen reader speech output from the 2D position of the user's cursor in order to facilitate richer spatial understanding of Web pages. Our formative study and web extension systems do not require knowledge of new tones and sounds, instead spatializing speech experienced through desktop screen readers as-is. Furthermore, we aim to run our systems on real websites to research the benefits and drawbacks of this type of a tool when placed in the context of a BLV user's typical desktop web browsing session.

3 FORMATIVE STUDY: ATTITUDES TOWARD SPATIAL INTERACTIONS IN SCREEN READERS

We recruited nine BLV participants, ages ranging from 22 to 65, to participate in a formative study. Our participants were either regular screen reader users or highly familiar with them. Table 1 shows a breakdown of their information. Participants were located in various USA time zones, so all studies were conducted remotely over Zoom, and lasted no more than 2 hours. Participants were compensated with a \$25 gift card for their time. Our study was approved by our institution's Institutional Review Board (IRB).

3.1 The SpaceNav System

We employed spatial audio output to keep users familiar with screen reader paradigms without introducing hardware to their setups. We were also keen on experimenting with spatialized input, as one of the constraints of working with screen readers is that the input mechanisms (e.g., Tab and Shift-Tab) make it difficult to control navigation spatially.

We created a screen reader experience named SpaceNav with two different configurations for users to explore web applications with: spatial input (which we will refer to as SpaceNav-Input), and spatial output (which we will refer to as SpaceNav-Output). SpaceNav-Input repurposes the four arrow keys (up, down, left, right) as input mechanisms for directional navigation of content, but also retains navigation using Tab and Shift+Tab so that users can navigate content in their preferred manner. SpaceNav-Input renders audio non-spatially, similar to existing screen readers. SpaceNav-Output extends SpaceNav-Input by incorporating spatialized audio output. The SpaceNav-Output configuration renders all screen reader speech output from a point sound source located at the 2D position on the user's screen.

We developed two different web applications using HTML, CSS, and TypeScript that integrated SpaceNav. The first web application, shown in Figure 1, was a mock news website designed to help us understand how users consume web content with spatial interactions. The second web application was a content creation application in which users created an "About Me" page, as seen in Figure 2. We used the Web Audio API to implement spatial audio for SpaceNav-Output. This makes use of stereo headphones to convey the effect of horizontal panning. To get the spatialized audio for each element in SpaceNav-Output, the midpoint position of each element was calculated and divided by the width/height of the screen for the x/y position respectively. We added changes in pitch [7, 11, 38] as

PID	Age	Gender	Left Eye Vision	Right Eye Vision	Onset	Hearing Impaired?
1	34	Male	No usable vision	No usable vision	Birth	No
2	49	Male	Total blindness	Total blindness	Birth	No
3	23	Male	Total blindness	Reduced acuity	Birth	No
4	25	Male	Total blindness	Total blindness	Birth	No
5	37	Male	Total blindness	Total blindness	Birth	Yes (right ear)
6	24	Male	Total blindness	Total blindness	Birth	No
7	28	Female	Some light perception	Some light perception	Birth	No
8	26	Male	Total blindness	Total blindness	Birth	Yes (both ears)
9	65	Female	No light perception	Total blindness	Age 57	No

Table 1: Self-reported demographics of our study participants.

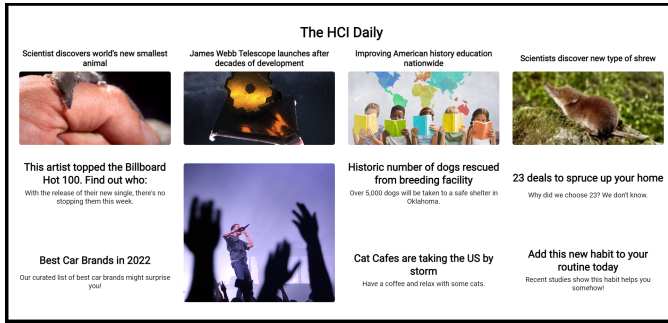


Figure 1: Version 1 of the news site, which users experienced with SpaceNav in our studies.

a redundant cue for representing vertical position. In our sites, we associated higher-pitched speech with elements that were visually higher on the page.

3.2 Study Procedure

Each user session consisted of three parts. First, participants used each of our screen reader configurations to navigate two different layouts of the news website — SpaceNav-Input for one of the layouts, and SpaceNav-Output for the other (with the layout ordering randomized across participants). Next, participants used SpaceNav-Input and SpaceNav-Output to create an About Me page (created pages by P6 shown in Figure 2). For both the news website and the About Me page creation, we asked users to explore the site at their own pace and vocalize their reactions to the system and interpretations of where specific elements were on the page. Last, we asked about participants' holistic experiences with the spatial interactions that they encountered. This included an overall comparison of SpaceNav-Input and SpaceNav-Output, as well as asking about which (if any) other scenarios they felt could benefit from spatial interactions. We used thematic coding [36] and a grounded theory approach [35] to analyze the qualitative results of our data.

4 FORMATIVE STUDY: RESULTS AND DISCUSSION

All nine participants preferred using the arrow keys as their input mechanism in both SpaceNav configurations instead of Tab and Shift+Tab. SpaceNav-Input benefited users primarily by helping

them better understand relationships between elements. P4 described website actions with his screen reader as feeling *"very basic when you have it presented in a very linear fashion"*. With SpaceNav-Input, he appreciated *"having the capacity to discern where things are again with respect to one another"*.

SpaceNav-Output showed us that screen reader users value having spatial feedback of the website as they were navigating it for both consumption and production scenarios. P5 said: *"I really liked the spatial audio...it helps me build a mind to picture of where things actually are located."* Participants also expressed a sense of increased independence as a result of feeling more immersed in and aware of the layout of a page using SpaceNav-Output.

Six participants felt SpaceNav-Output reduced cognitive load. They also found it easier to orient themselves on a given web page. Participants expressed excitement at not needing to rely on memory, counting, or understanding coordinates of their location to figure out where they were at any given time. Furthermore, participants commented on how they could envision better communication between themselves and sighted users/visual learners: *"It's cool to visualize because I can't tell you how many times if I need help with something... a sighted person references 'oh on the top left of the screen'; I don't know what's on the top left of the screen."*— P7

Alongside benefits, participants also expressed some drawbacks of the SpaceNav system. They were unsure how directional arrow key navigation would integrate well with existing screen reader keyboard mappings for the up, down, left, right arrow keys. Users also discussed how losing categorical grouping associations by navigating directionally could make it hard for them to be confident in the content they have consumed.

Three participants experienced increased cognitive load when using SpaceNav-Output. Since this finding was at odds with other participants saying that SpaceNav-Output reduced their cognitive load, we probed further to understand the discrepancy. Participants who felt added cognitive load were very familiar with screen readers (they were lifelong screen reader users). For them, having to interpret spatialized audio was new and felt unnecessary. From this, we hypothesize there might be a difference in usefulness of spatial interactions between lifelong screen reader users and users who have learned to use screen readers at a later stage in life. We plan to explore this in our future study.

In our interviews, participants identified many scenarios that could benefit from spatial interactions. These included gaming, route-finding using spatial interactions to evaluate multiple route



Figure 2: P6’s About Me pages - the page on the left was built using SpaceNav-Input, and the page on the right was built using SpaceNav-Output. In explaining why the two About Me pages were different, P6 discussed how spatial audio made a difference in helping him realize that he did not want images to be in between headings and the associated text as it interrupted the flow of text.

options, classroom use such as interpreting a binary tree for a computer science course, and giving live presentations to groups consisting of sighted and BLV people.

5 LIMITATIONS AND BROWSER EXTENSION DEVELOPMENT FOR FUTURE WORK

Our SpaceNav implementation had several limitations: it was built to work only on our custom web applications, it did not handle multi-page websites that would involve scrolling, and participants were concerned that SpaceNav would not extend to pages that had more nuanced, non-grid layouts. As a result, we are currently developing a Google Chrome extension which we call SpaceNav-Chrome to enable a spatialized audio screen reader experience on any website, including popular websites such as Gmail and Reddit.

For SpaceNav-Chrome, we are extending our spatial audio system to support web pages that scroll rather than single-page websites only. To do so, we are adding a scroll indicator in the form of a spatialized beep to alert users whenever the page scrolls and UI elements “jump” to a new vertical position. We are also in the process of recruiting participants for a user study to test SpaceNav-Chrome on live websites. We plan to have participants compare using their current screen reader versus SpaceNav-Chrome on different types of sites that they themselves frequently use, using semi-structured interviews to gauge benefits and drawbacks of the browser extension. Through this study, we plan to establish guidelines for incorporating spatial interactions into BLV users’ desktop screen reader web workflows to maximize the benefit of spatial interaction while avoiding the drawbacks that we observed in our formative study.

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